



HABIB UNIVERSITY

DHANANI SCHOOL OF SCIENCE AND ENGINEERING

EE251/CE252	Spring'26	Quiz 4 SOLUTION	Marks 30 Weight 4%	April 13, 2026
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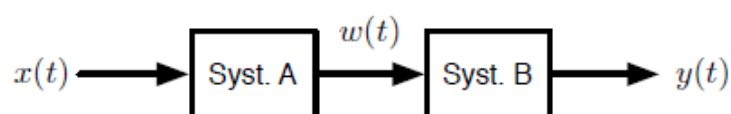
Important Notes

Clearly write your name and student ID on top.

CLO-4	Analyze CT signals and systems in the transform domain using Laplace, Fourier transforms and Fourier Series.	PLO-2 (Problem Analysis)	Cognitive Level-4 (Analyzing)
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Problem 1 [30 pts]

Consider the system shown below



With

$$\frac{dw(t)}{dt} + w(t) = x(t), \quad \frac{dy(t)}{dt} + 2y(t) = w(t)$$

- (a) Assuming zero initial conditions, find the transfer functions of System A and System B
- (b) Find the impulse response of System A and System B.
- (c) Check whether System A and System B are stable or not. Justify your answer in both time and frequency domain.
- (d) Check whether System A and System B are [stable] invertible or not.
- (e) Given $X(s) = 2s^2 + 5$, find $y(t)$.

(f) Suppose you would like to add a third system (system C) next to system B such that the combined systems simply scales an input by a factor of 10, i.e., input $x(t)$ becomes $10x(t)$. Find the transfer function $H_C(s)$.

(Hint: By Laplace Transform properties, $\frac{dx(t)}{dt}$ has Laplace transform $sX(s) - x(0)$).

TABLE 4.1 Select (Unilateral) Laplace Transform Pairs

No.	$x(t)$	$X(s)$
1	$\delta(t)$	1
2	$u(t)$	$\frac{1}{s}$
3	$tu(t)$	$\frac{1}{s^2}$
4	$t^n u(t)$	$\frac{n!}{s^{n+1}}$
5	$e^{\lambda t} u(t)$	$\frac{1}{s - \lambda}$

Solution:

(a) We have

$$(s + 1)W(s) = X(s)$$

$$(s + 2)Y(s) = W(s)$$

Leading to System A transfer function $H_A(s) = \frac{1}{s+1}$, and System B transfer function

$$H_B(s) = \frac{1}{s+2}.$$

(b) Since impulse response is the inverse transform of transfer function, we use the Laplace Transform Table to get

$$h_A(t) = e^{-t} u(t)$$

$$h_B(t) = e^{-2t} u(t)$$

(c) Both systems are stable since their poles ($s = -1, s = -2$) are in the Left Half Complex Plane. In time domain, we see that $\int |h_A(t)| < \infty$ and $\int |h_B(t)| < \infty$ (since both are exponentially decaying functions), which satisfies the condition for stability in time domain.

(d) Both systems have no zeros in the right half plane. Therefore, both are stable invertible.

(e) We have

$$Y(s) = H_A(s)H_B(s)X(s) = \frac{2s^2 + 5}{(s+1)(s+2)}$$

$$= 2 + \frac{k_1}{s+1} + \frac{k_2}{s+2}$$

PFE gives $Y(s)$ as

$$= 2 + \frac{7}{s+1} - \frac{13}{s+2}$$

Inverse Laplace gives $y(t)$

$$= 2\delta(t) + (7e^{-t} - 13e^{-2t})u(t)$$

(f) Setting $H_C(s) = 10(s+1)(s+2)$ gives the desired result.